



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	ADVANCED ORGANIC CHEMISTRY				
Type of Course	ELECTIVE IN MAJOR				
Semester	VIII				
Academic Level	400 – 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-		60
Pre-requisites	1. Prior knowledge of chemical bonding and structure of molecules 2. Understanding the reaction mechanism of different reactions and rearrangements.				
Course Summary	This course covers reactions, synthesis planning, supramolecular chemistry, and drug design				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To understand and apply named reactions and reagents in organic synthesis	U	P	Quizzes /Seminar
CO2	To formulate synthetic strategies employing synthons, protecting groups, and reagents	An	C	Discussion/ Assignment
CO3	To conceptualize non-covalent interactions and applications in supramolecular chemistry	An	C	Seminar / Discussion
CO4	To Comprehend the principles of drug design and understand the	Ap	F	Discussion/Seminar /Assignment

	stages of drug development			
CO5	To Rationalize the mechanisms of redox reactions and substitution reactions involved in organic synthesis	Ap	An	Seminar/discussion
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Named Reactions and Reagents		9	18
	1	Reformatsky reaction, Baeyer-Villiger oxidation, Beckman rearrangement,	3	
	2	Benzilic acid rearrangement, Benzoin Condensation, Claisen rearrangement,	2	
	3	Clemmenson reduction, Dies-alder reaction, Knoevenagel condensation,	2	
	4	Pictet-Spengler reaction, Strecker amino acid synthesis, Simmons-Smith reaction,	2	
II	Synthesis and Synthetic Planning		13	26
	5	Target molecules, Synthons, synthetic equivalents (anionic and cationic)	1	
	6	Disconnection	1	
	7	Protecting groups-alcohols (Bn, PMB, Ac, TBS) explain the concept with a reaction	2	
	8	Protecting groups-aldehydes (cyclic acetal) explain the concept with a reaction	2	
	9	Protecting groups-amines (Bz, Boc, Fmoc) explain the concept with a reaction	2	
	10	Protecting groups-acids (Me and t-Bu ester) explain the concept	2	

		with a reaction		
	11	Convergent and linear synthesis (demonstrate with an example)	1	
	12	Functional inter-group conversions (Any one example), retrosynthesis of chalcones.	2	
III	Supramolecular Chemistry		13	26
	13	Various types of non-covalent interactions (H bonding, van der Waals interactions, cation-pi interaction, pi-pi stacking) bonding and applications of addition compounds,	4	
	14	Crown ethers	2	
	15	Cyclodextrins	2	
	16	Cryptands, catenanes and rotaxanes.	2	
	17	Importance of supramolecular chemistry in living systems.	3	
IV	Principles of Drug Design		13	28
	18	Introduction to medicinal chemistry	2	
	19	Therapeutic index, solubility, Intermolecular binding forces in drug target interactions- (electrostatic, H bonding, van der Waals, dipole-dipole).	3	
	20	Introduction to various drug targets; Proteins- Enzymes- Receptors- their roles, neurotransmitters, receptor activation and regulation.	3	
	21	Introduction to Pharmacodynamics and pharmacokinetics (ADME); affinity and efficacy.	3	
	22	Types of drugs, stages of drug development (basic concept only).	2	
V	Open ended		12	
	Modern Techniques in Organic Synthesis/Asymmetric Synthesis/Total synthesis			

References

1. Organic Chemistry, by Jonathan Clayden, Nick Greeves, Stuart Warren, Oxford University Press
2. March's Advanced Organic Chemistry: Reactions, Mechanisms and Structure, Michael B. Smith, Wiley
3. Organic Chemistry Paperback, Francis Carey, McGraw-Hill Education.
4. Advanced Organic Chemistry, Jerry March.
5. Advanced Organic Chemistry, David E. Lewis, Oxford Univ Pr; Illustrated edition.
6. Principles of Organic Synthesis, R. O. C. Norman & J. M. Coxon, 3rd Ed., CRC Press, 2000.

7. The Art of Writing Reasonable Reaction Mechanisms, R. B. Grossman, 3rd Ed., Springer, 2019
8. Name reactions J J Li, Springer.
9. Medicinal Chemistry, Sriram and Yogeeswari, Pearson Education India; 2nd edition.
10. An Introduction to Medicinal Chemistry, Graham Patrick, Oxford University Press; International edition.
11. Organic Chemistry, Morrison Boyd & Bhattacharjee, Pearson Education India; 7th edition

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1						3		1		1		2
CO 2	2						3		2		3		3
CO 3	2						3		2		3		3
CO 4					2	1	3		2		3		3
CO 5				2	1	3	3		2		3		3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Pr actical Exam	Assignmen t /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓